REPORT : Waste Segregation model , Automating garbag classification...

model overview

- model based on Resnet50 Architecture
- train accuracy is about 92%(5 epochs) and test accuracy is about 90%
- Model summery(Resnet50)

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Model: "sequential_1"
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 Layer (type)                Output Shape              Param #
=================================================================
 resnet50v2 (Functional)     (None, 7, 7, 2048)        23564800

 dropout_2 (Dropout)         (None, 7, 7, 2048)        0

 batch_normalization_2 (Bat  (None, 7, 7, 2048)        8192
 chNormalization)

 flatten_1 (Flatten)         (None, 100352)            0

 dense_2 (Dense)             (None, 64)                6422592

 batch_normalization_3 (Bat  (None, 64)                256
 chNormalization)

 dropout_3 (Dropout)         (None, 64)                0

 dense_3 (Dense)             (None, 12)                780

=================================================================
Total params: 29996620 (114.43 MB)
Trainable params: 22779852 (86.90 MB)
Non-trainable params: 7216768 (27.53 MB)
_________________________________________________________________
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- Dataset having 12 different classes [Battery , biological , paper , cardboard , white-glass , brownglass , cloths, plastic , trash , shoes , metal , greenglass]

FUTURE WORK AND CHALLENGES :

- try more images or data agumentaion to handle imbalaned dataset
- compare VGG16 , Resnet50, MobileNet , Custom CNN with Data agumentaion
- Try to implement with yolo , for faster detection
- Certain waste items, such as plastic and metal, may look similar in certain conditions (e.g., reflective surfaces, translucent materials), leading to misclassification.
- Indian moms pack food in black polythene bags, making it challenging for the model to predict accurately.
- Some waste types, such as plastic, are more common than others like metal or glass, leading to an unbalanced dataset that biases the model(since F1 score is 0.16).

Garbage origin (IITBHU)

CURRENT DISPOSEL METHOD

- The existing waste management system at IIT(BHU) primarily involves collection and transporting waste to landfills.
- There is no facility of treatment and recycling process within the campus. Hence, all the waste collected is transported to the dumping ground

WASTE AUDIT..

- **Waste Composition Analysis**: Determine the types and quantities of waste generated on campus (organic, plastic, paper, etc.).
- **Identify Waste Sources**: Pinpoint where most waste is generated (e.g., cafeterias, hostels, labs) to tailor the solution effectively.
- **Analysis**: Use the collected data to prioritize high-volume waste sources and develop targeted waste management solutions.

Table 3: Two-week data of quantification of solid waste generated in different hostels. (Under the category of Hostel Sweeping Waste) in kg.

S.No.	Name of Hostel	in kg	S.No.	Name of Hostel	in kg
1	C. V. Raman	**254.5**	*9*	Vishwakarma	**253.4**
2	Morvi	**233.34**	*10*	Gandhi Smriti	**168.2**
3	Dhanrajgiri	**308.5**	*11*	New Girls GSMC Extn.	**169.1**
4	Rajputana	**393.88**	*12*	Saluja Hostel	**83.2**
5	Limbdi	**283.2**	*13*	G.R.T Apartments	**199.3**
6	S.C. De	**162.3**	*14*	S.N. Bose	**253.1**
7	Visvesvaraya	**330.1**	*15*	S. Ramanujan	**114**
8	Vivekanand	**269.6**	*16*	Aryabhatta	**349.7**
				Grand Total	**3825.42**

TRACING THE WASTE FOOTPRINT OF IIT-BHU

FROM VT : A huge amount of biodegradable waste is generated every day from the Vishwanath temple area in the campus. Total waste is collected as a sample and sorted out for studying its biodegradable and non-biodegradable components. Further, 10 percent of temple waste is segregated to estimate the content of flower and leaf compositions in biodegradable waste generated from VT

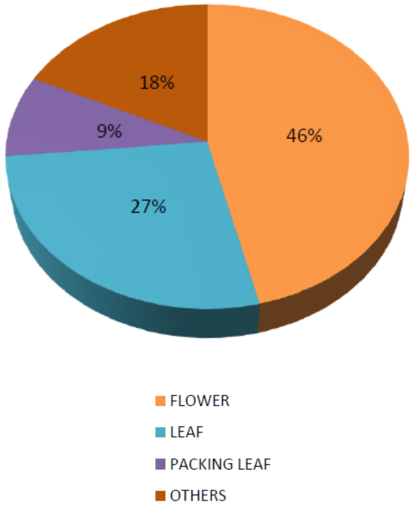


Fig: Various components in MSW at VT per day

FROM FRUIT SHOPS : There are three fruit shops on campus, and their daily waste, mainly fruit peels, is disposed on the roadside. Since the waste is entirely organic, it doesn't require segregation and can be directly used for cattle feed, composting, or in a biogas plant.

FROM HOSTEL MESS : The hostel mess generates about 1061 kg (or 1 ton) of waste per day, averaging 0.21 kg per person. This waste is collected from bins outside the mess and is entirely used as cattle feed by local milkmen, so there’s no food waste accumulation.

FROM HOSTEL AND CANTEEN : At IIT (BHU), waste from more than 14 canteens was sampled and sorted daily to determine the quantity of different components. The study focused on calculating the percentage of compostable materials in the waste.

To assess solid waste generation in hostels, waste from four hostels was weighed and studied over two weeks. This analysis focused on identifying the physical characteristics of the waste to help develop strategies for reducing, recycling, and reusing waste.

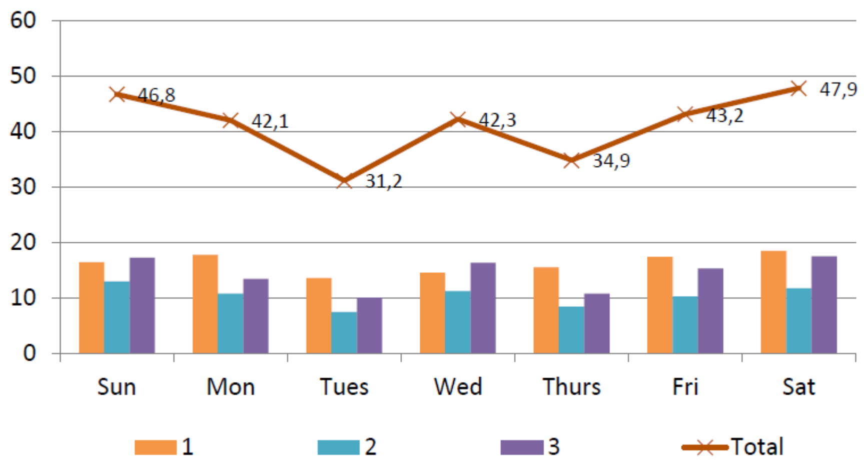


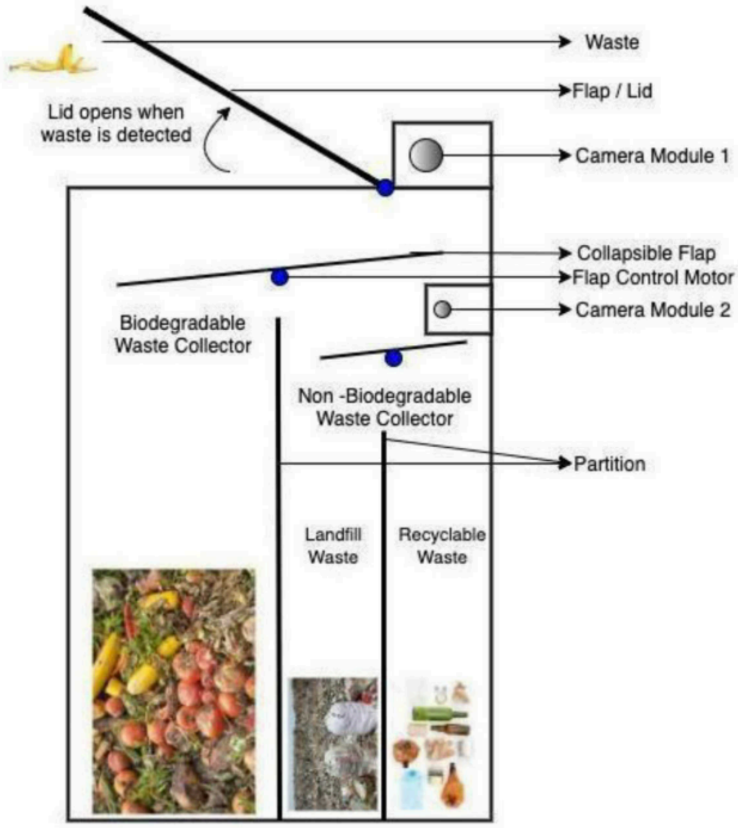
Fig: One-week data of solid waste of fruit shops in IIT (BHU)

AUTOMATION FOOTPRINT

OVERVIEW : Increase in population, progress in technological advancement, urbanisation and development have resulted in an increase growth in consumer product. And with this progress comes a price; generation of waste. In a recent survey conducted by the World Bank, about 1.3 billion tons of waste is generated each year. The number is expected to reach 2.2 billion by 2025. Waste management is a problem faced by a lot of societies and communities. The amount of waste generated is far more than the waste recycled. Improper waste management has led to increase in cost of recycling and more working hours. This leads to overcapacity of dumping grounds and landfills, littered waste around the city. These environments act as a breeding ground for diseases.

WORKING

- An ultrasonic sensor detects when trash is near the smart bin's lid, automatically opening it to allow users to place their waste on the first collapsible flap for easy disposal.
- The smart bin autonomously segregates waste into biodegradable and non-biodegradable categories, minimizing user interaction and ensuring efficient waste management while promoting environmental sustainability.
- Advanced sensors, including a Raspberry Pi camera, infrared sensor, and moisture sensor, work together to accurately classify waste, enhancing the efficiency of the segregation process for improved recycling efforts.
- Servo motors control the movement of collapsible flaps, tilting them based on the classification pulse received, ensuring trash is directed into the appropriate compartments without requiring further user input.
- Non-biodegradable waste is further analyzed for classification into landfill or recyclable materials using object detection algorithms, ensuring that waste is sorted effectively to maximize recycling potential.



- Try to build protype for IITBHU campus , for effective segregation and energy conversion

ENERGY CONVERSION

priliminary result..

Anaeoribic digestion

- Assumed average biogas yield is **0.5 m³/kg** of biodegradable waste, influenced by composition, temperature, retention time, and the anaerobic digestion process used.
- From **409 kg** of biodegradable waste, using anaerobic digestion, you could generate approximately **250 to 375 kWh** of energy, depending on the biogas yield.

Fermentation

- Assumed waste is mainly organic (e.g., food, agricultural residues) for anaerobic digestion, yielding 0.5 m³ of biogas/kg, varying with material type and fermentation conditions.
- Biogas Yield: **100 to 200 m³/ton** and Energy Content: **20 to 25 MJ/m³** and Total Energy: **~227.2 to 567.6 kWh** from **409 kg** of biodegradable waste

Incineration

- Assumed average calorific value is **8 MJ/kg** for mixed non-biodegradable waste, with **85%** incineration efficiency for heat-to-electricity conversion, varying based on waste composition.
- Total Energy from **411 kg** of Non-Biodegradable Waste via Incineration is about**~ 1,142 kWh to 2,283 kWh.**

Pyrolysis

- It is assumed that the pyrolysis of **411 kg** of non-biodegradable waste yields approximately **50% bio-oil, 10% gas, and 20% char**, reflecting typical values for such waste types.
- Total Energy is about **~1,713 kWh to 3,650 kWh** from **411 kg** of non-biodegradable waste via pyrolysis.

future work : work more on enerygy conversion techniques